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Introduction to Databases

Database Systems & Information Modelling
INFO90002

Week 1 – Database basics
Dr Tanya Linden
David Eccles

Database Systems





Learning Objectives

Difference between data and information

Unstructured/Semi-structured/Structured data

Database and DBMS – not the same!

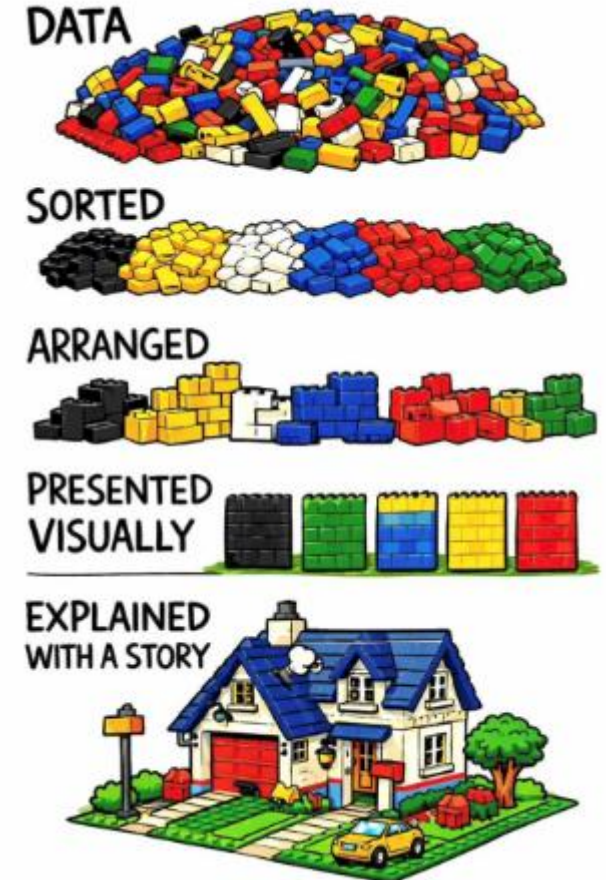
Relational DBMS

Client server architecture

Database development lifecycle and importance of database design

Context: software and data

- Computer systems consist of *software* working to process *data*.
- You will learn about creating software and algorithms in other subjects (COMP90041)
- This subject is about *data*.
- Focus away from *computation* to *data*
Processed data = information



Source: Instagram of BusinessInfographics

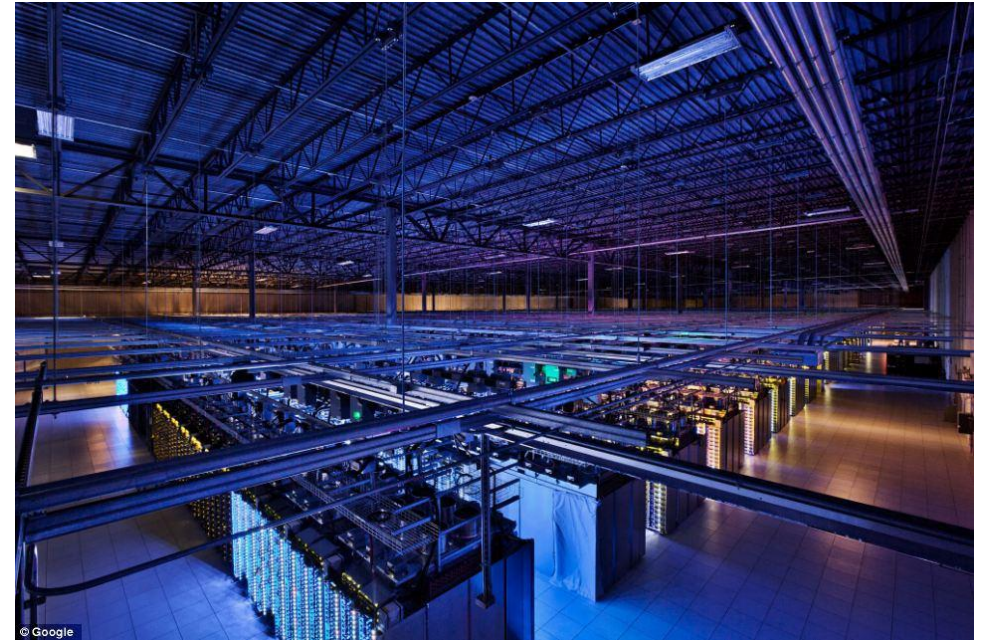
The Modern Data Challenge

Modern organisations need to store and retrieve large amounts of data

Data can be divided into three major **categories**

- **Structured Data**
- **Semi-Structured Data**
- **Unstructured Data**

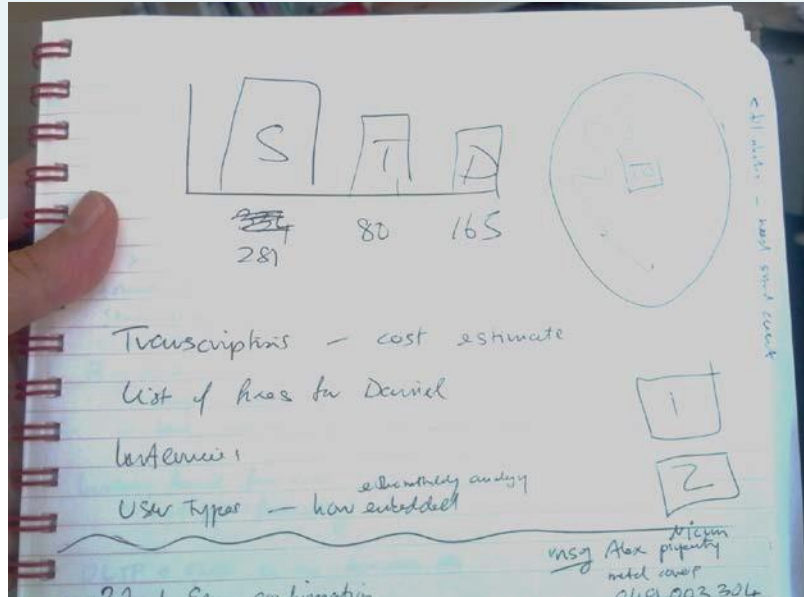
Structured Data is typically used by Relational Database Management Systems (RDBMSs) such as Access, Oracle, SQL Server, MySQL.



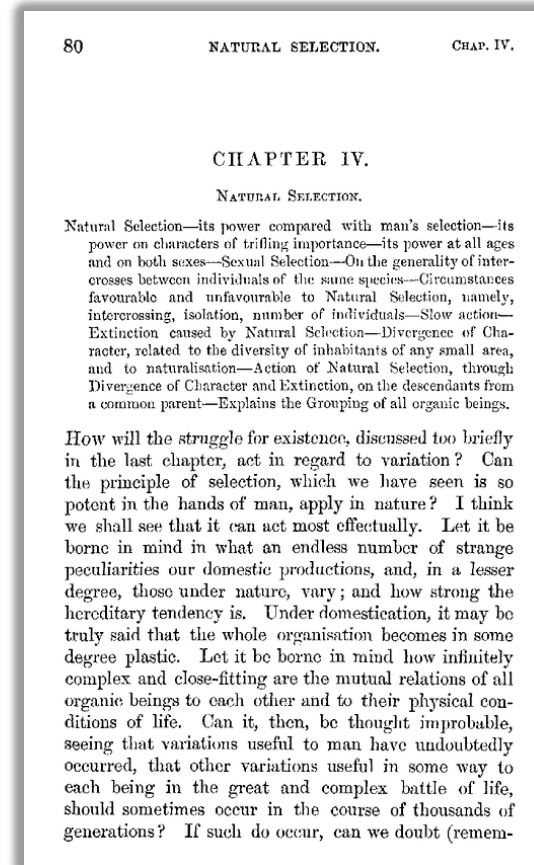
Inside Google's Giant Data Centers

<https://markosun.wordpress.com/2012/10/21/inside-googles-giant-data-centers/>

Unstructured data



	A	B	C	D	E
1	My Holiday Budget ----->				
2					
3	Flights	\$10,000.00			
4	Accommodation	\$5,000.00	<- can this be reduced?		
5	Food	\$1,000.00			
6					
7		\$16,000.00	TOTAL		
8					
9					
10					
11					



- handwritten notes
- printed books
- spreadsheets (semi-structured?) etc.



Unstructured Data

Unstructured data is not organised in a pre-defined manner.

The organisation **does not know the format**, nor the content of the data in advance.

Consider data sourced from **social media, email**, etc. The contents are unpredictable.

- The data may contain text, audio, video, links, images. One item may include many data about many organisational functions.

How do organisations store such data so that it can be retrieved, collated, analysed?



Semi-Structured Data

Semi-structured data is information that doesn't match the requirements of a relational database.

The data is organised / arranged that makes it easier to analyse.

Examples of semi-structured data include **XML** documents and **NoSQL** databases.

We will briefly deal with the topic of semi-structured data in future weeks



Structured Data

Relational Database Management Systems require data to be stored in a very structured way.

These systems deal with data that has a **repetitive pattern or format**.

Consider **Student** data stored by a University. While every student is different, the university wants to store data in the same format for every student. Data Types are also specified for each piece of information.

- | | |
|-------------------------------|--|
| • Student ID – Numeric/Digits | HomeAddress - Alphanumeric |
| • Student Name – Alpha | PhoneNo – Numeric or Digits+space+brackets |
| • Gender – Alpha | NextOfKin – Alpha |
| • DateOfBirth – Date | |



Why use a DBMS?

DBMS = **D**ata**B**ase **M**anagement **S**ystem

Data independence

- Applications should not be exposed to data representation and storage
- DBMS provides an abstract view that hides representation and storage

Efficient access

- More efficient data storage and retrieval than flat files

Data integrity and security

- DBMS enforces data integrity constraints, access controls and govern user access
- Not reliant on just the operating system



Why use a DBMS?

Uniform data administration

- Specialist skills in data management and administration
- Layer of expertise reduces risk to data and data owners

Concurrent access and crash recovery

- Schedules concurrent access. Protects data from system failures

Datasets increasing in diversity and volume

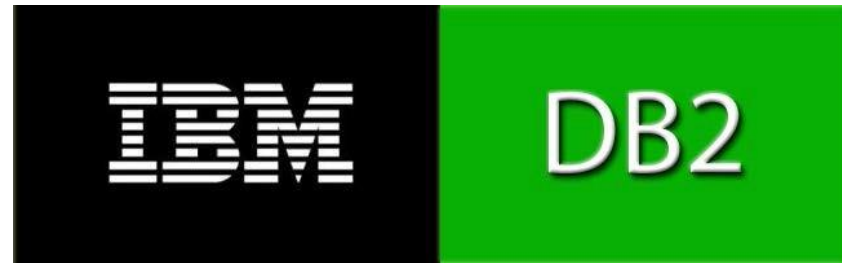
Data Independence

- Logical data independence: the ability to modify the logical schema without causing application programs to be rewritten (e.g. Add/Modify/Delete a new attribute, entity or relationship)
- Physical data independence: e.g. using a new storage (Cloud instead of local server Hard Drive; changes to compression techniques)

Reduced Application Development time*

Relational Databases

- The first relational databases from Oracle and IBM appear around 1980



- Others appear later





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Client- Server Architecture

Introduction to MySQL Server
and MySQL Workbench

RDBMS and Database Servers

A database server is a computer that is **networked to other computers**

The database server **stores databases**

Users on the network **can access the data** stored in the databases

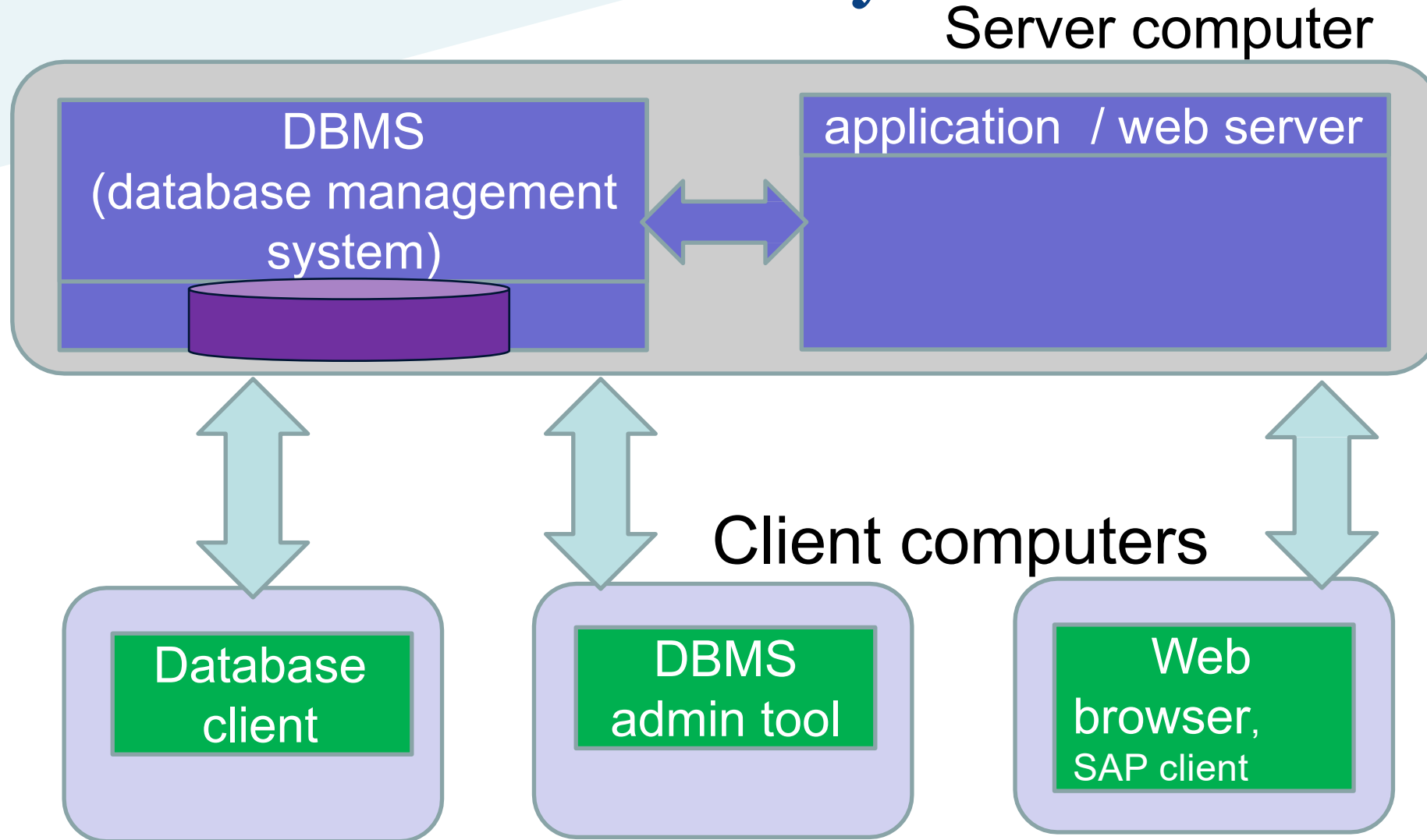
There is only **one copy** of the data (excluding backups, etc.)



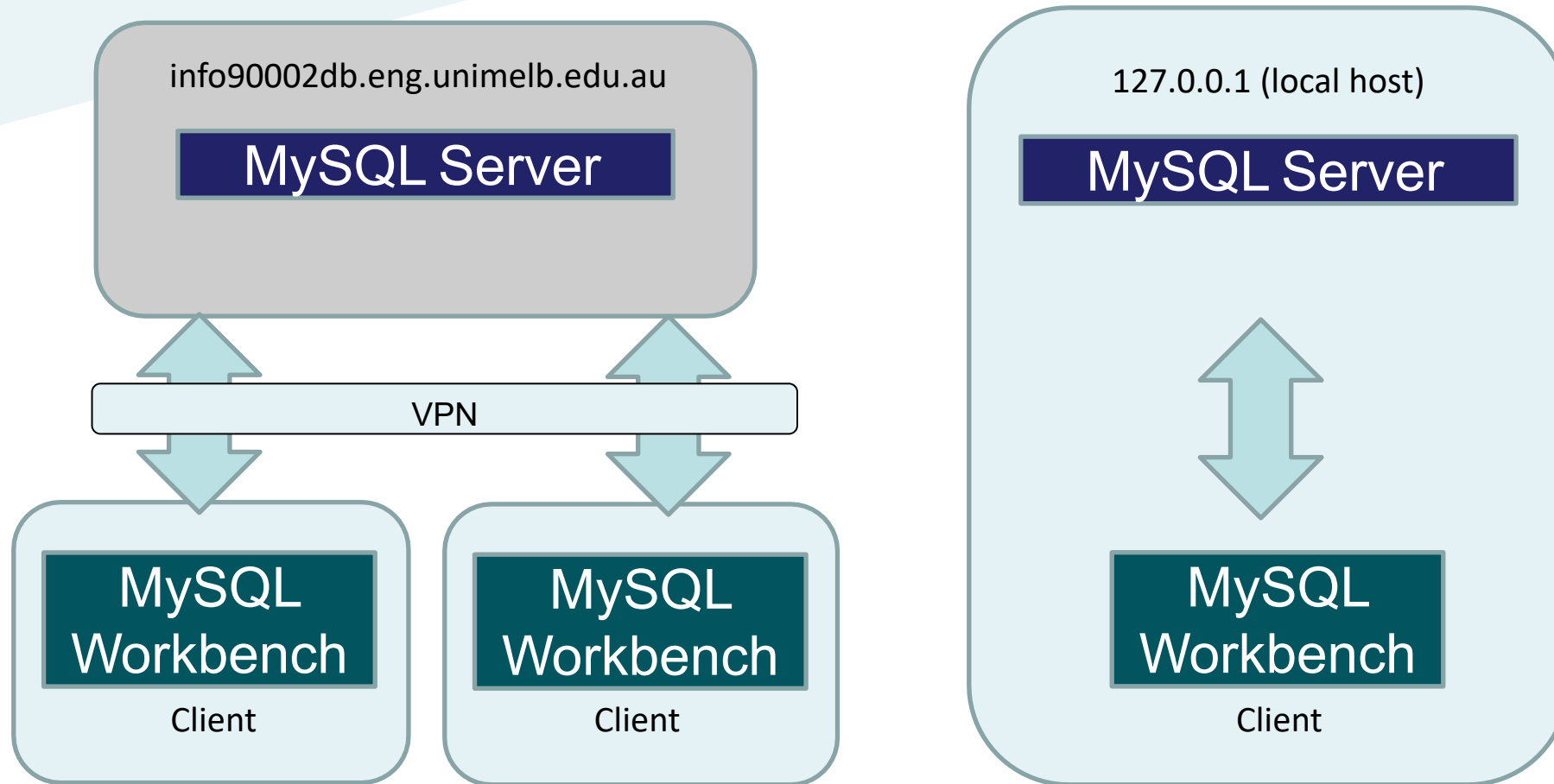
RDBMSs allow **multiple users on the network to update data** in database tables.

- Many people can check the price of product 20
- Many people may choose to enroll in INFO90002 simultaneously
- Many users may purchase tickets for a flight at the same time

Client Server – In Industry



... in teaching (and development)



Labs or own laptop

Your PC



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Relational databases

Basic concepts





Relational Data Model

RDBMSs are based on the Relational Data Model

- Developed by Ted Codd in 1970.
- Data is represented in the form of two-dimensional **tables**.

Each two-dimensional table has the following properties:

- A set of uniquely named **Columns / Attributes**
- A list of unnamed/unnumbered **Rows**
- The **order** of the rows is **irrelevant**.

A **Row** consists of a sequence of **Attributes**

- **One cell** for each Attribute
- Only **one value per cell** is allowed.

All Relational Database Management Systems are **based on** the Relational Data Model.

Relational Data Model (cont.)

Table

Uniquely named columns called Attributes

Attribute

Unnamed and
unnumbered Rows

Row →

<u>ID</u>	StName	Degree	Enrolled
1001	Josh Blogs	615	20/01/2024
1004	Emma Lewis	308	03/02/2023
1006	Dave Rossi	234	10/07/2024

The order of the rows is irrelevant.

Only one value per cell allowed.



Relational Databases

A relational database is a collection of related tables

The **relational model's** foundation is a mathematical concept known as a relation

- A **relation** is a two-dimensional structure composed of intersecting rows and columns
 - Imagine it as a table
- Each row in a relation is called a **tuple** and each column represents an **attribute**

Example:

- **Student Table** (stores data about students)
- **Subject Table** (stores data about university subjects)
- **Enrolment Table** (stores data about a student's enrolment into a specific subject in a specified year and semester)

RDBMS

A RDBMS is a collection of programs that allow developers / users to store and retrieve data from relational databases

It allows users to perform CRUD (create, read, update and delete) operations on data in the tables, e.g.

- **Create** a student record
- **Retrieve** the student's details
- **Update** the student's details
- **Delete** the student from a table



Setting up a RDBMS

Tables

- Follow a 2-dimensional structure
- Each row of data is identified by a unique Primary Key
- No duplicates, e.g. Student ID

Constraints (based on business rules) can be added to validate data

- Student ID is correct length
- Student type is PG or UG (postgrad or undergraduate)
- Student is enrolled in a subject that is actually offered in the current semester

Case Study - Overview

Let's consider a business called **Pizza OnLine**

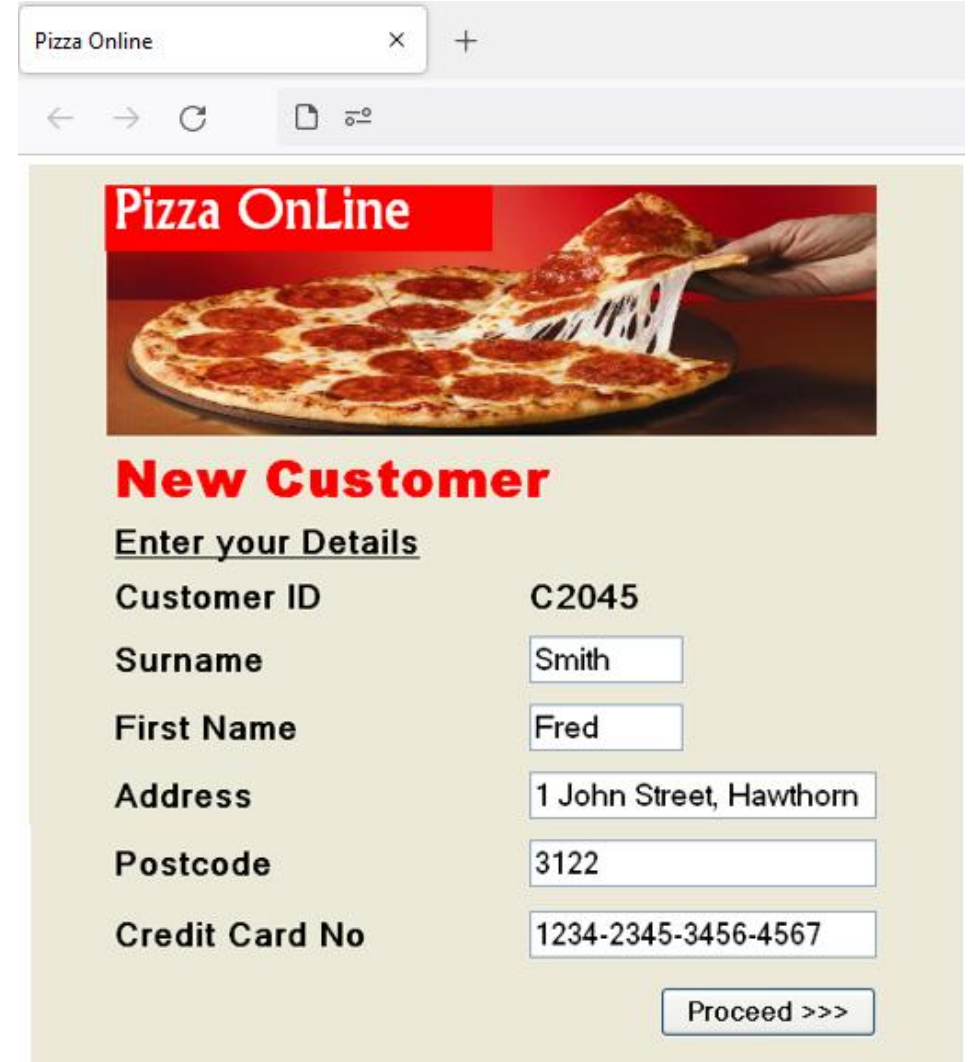
- It allows customers to **order pizzas** via the online portal.
- Customers have pizzas delivered.
- Only people who register with a credit card number can be an on-line customer.



Case Study - Customer Details

Customers need to supply personal details:

- Name
- Address
- Credit Card No



The screenshot shows a web browser window with the title "Pizza Online". The page features a header image of a pepperoni pizza with a hand lifting a slice. Below the image, the text "New Customer" is displayed in red, followed by the instruction "Enter your Details". The form contains several input fields for customer information, including Customer ID, Surname, First Name, Address, Postcode, and Credit Card No. A "Proceed >>>" button is located at the bottom right of the form.

Customer ID	C2045
Surname	Smith
First Name	Fred
Address	1 John Street, Hawthorn
Postcode	3122
Credit Card No	1234-2345-3456-4567

Case Study – Order Details

The Order Form

- This is **One order**.
- The customer may add **multiple items** and **different quantities**.

Pizza Online

←

→

↻

📄

🔍



Order Form

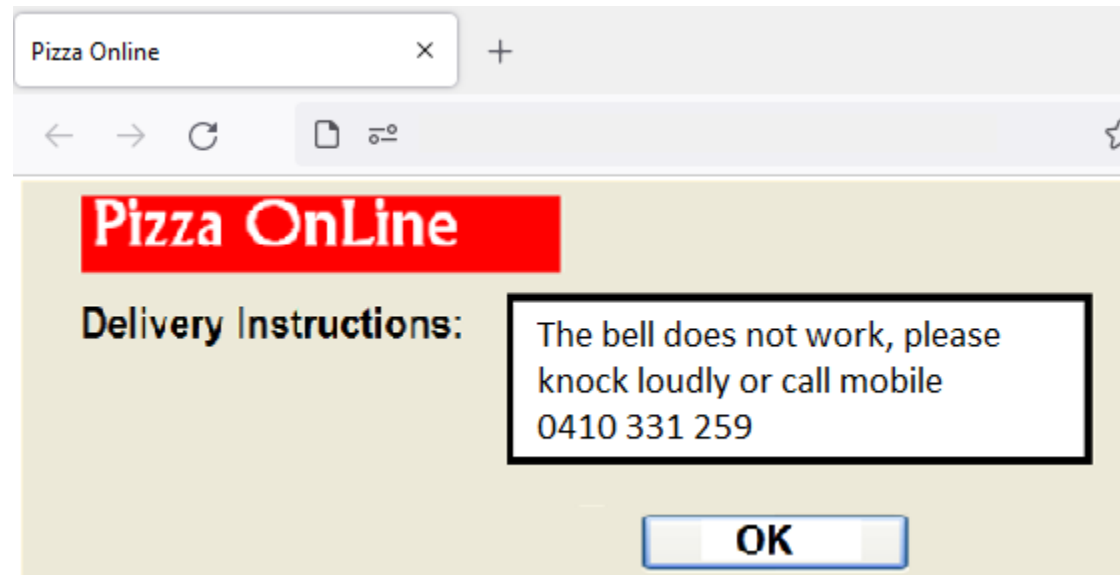
Choose Quantity of each item Required

	<u>Item</u>	<u>Price</u>	<u>Qty</u>
P1	Mario's Supreme Pizza	\$6.95	2
P2	Vegetarian Pizza	\$6.95	0
P3	Hawaiian Pizza	\$6.95	1
P4	Hot 'n' Spicey Pizza	\$6.95	1
B1	Garlic Bread	\$4.95	4
B2	Herb Bread	\$4.95	0
D1	2 Litre Cola	\$2.50	2
D2	2 Litre Lemonade	\$2.50	0

Proceed >>>

Case Study - Customer Details

Add any **delivery instructions** that are required for the order:



Case Study – The completed Order

This Order is for
Customer **C2045**.

Order Date and Time is
Sept 1, 2020 17:35

This order has special
delivery instructions

The order is for many
different Items.

Each item ordered
has a **Quantity** value



Pizza Online

Pizza OnLine
Confirm Your Order

Customer ID C2045 Date: Sept 1, 2020
Time: 17:35

Customer
Name Fred Smith
Address 1 John Street, Hawthorn, 3122

Delivery Instructions: The bell does not work, please knock
loudly or call mobile 0410 331 259

Item		Price	Qty
P1	Mario's Supreme Pizza	\$6.95	2
P3	Hawaiian Pizza	\$6.95	1
P4	Hot 'n' Spicy Pizza	\$6.95	1
B1	Garlic Bread	\$4.95	4
D1	2 Litre Cola	\$2.50	2
Total		\$59.55	

Confirm

Pizza Online Database

Customer ID	Surname	First name	Address	Postcode	Credit card
C2045	Smith	Fred	1 John St, Hawthorn	3122	1234234534564567

Order No	Date	Time	Delivery Instructions	Total	Code	Pizza name	Price	Quantity
3224	1/09/2021	17:35	The bell does not work,	59.55	P1	Mario's Supreme Pizza	6.95	2
3224	1/09/2021	17:35	The bell does not work,	59.55	P3	Hawaiian Pizza	6.95	1
3224	1/09/2021	17:35	The bell does not work,	59.55	P4	Hot 'n' spicy Pizza	6.95	1

Problems:

- Order details repeated for every pizza ordered
- How do we link the order to the customer?

Tables in the Pizza Online Database

Customer ID	Surname	First name	Address	Postcode	Credit card
C2045	Smith	Fred	1 John St, Hawthorn	3122	1234234534564567
C2048	Nguyen	Vincent	2/7 Oak Ave, Altona	3018	4554123423457899
C2146	Davis	Liz	32 Lyle St, Toorak	3142	4564564578970022

Order No	Line Item	Quantity
3224	P1	2
3224	P3	1
3224	P4	1
3224	B1	4
3224	D1	2

Code	Pizza name	Price
P1	Mario's Supreme Pizza	6.95
P2	Vegetarian Pizza	6.95
P3	Hawaiian Pizza	6.95
P4	Hot 'n' spicy Pizza	6.95
B1	Garlic Bread	4.95
B2	Herb Bread	4.95
D1	2 Litre Cola	2.50
D2	2 Litre Lemonade	2.50

Order No	Customer ID	Date	Time	Delivery Instructions	Total
3224	C2045	1/09/2021	17:35	The bell does not work,	59.55

Manipulating Tables Structure

Customer ID	Surname	First name	Address	Postcode	Credit card
C2045	Smith	Fred	1 John St, Hawthorn	3122	1234234534564567
C2048	Nguyen	Vincent	2/7 Oak Ave, Altona	3018	4554123423457899
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Code	Pizza name	Price
P1	Mario's Supreme Pizza	6.95
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B1	Garlic Bread	4.95
B2	Herb Bread	4.95
D1	2 Litre Cola	2.50
D2	2 Litre Lemonade	2.50

Working with tables is like working with files
– there are 4 things you can do:

CREATE a table

DROP (i.e. delete) a table

ALTER a table (e.g. add a column)

RENAME a table

Manipulating Table Contents

Customer ID	Surname	First name	Address	Postcode	Credit card
C2045	Smith	Fred	1 John St, Hawthorn	3122	1234234534564567
C2048	Nguyen	Vincent	2/7 Oak Ave, Altona	3018	4554123423457899
C2146	Davis	Liz	32 Lyle St, Toorak	3142	4564564578970022

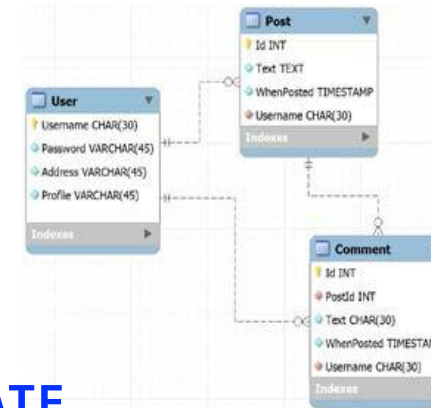
Code	Pizza name	Price
P1	Mario's Supreme Pizza	6.95
P2	Vegetarian Pizza	6.95
P3	Hawaiian Pizza	6.95
P4	Hot 'n' spicy Pizza	6.95
B1	Garlic Bread	4.95
B2	Herb Bread	4.95
D1	2 Litre Cola	2.50
D2	2 Litre Lemonade	2.50

For each table you need to be able to:

- SELECT**, or read, data from the table
- INSERT** new rows into the table
- DELETE** existing rows from the table
- UPDATE** existing rows in the table

Database lifecycle

- Design the database
 - data modelling, E-R diagrams
- Implement the database
 - data definition language DDL
- Data access / programming
 - data manipulation language DML
- Database administration
 - data control language DCL



- CREATE
- DROP
- ALTER
- RENAME

- SELECT
- INSERT
- UPDATE
- DELETE

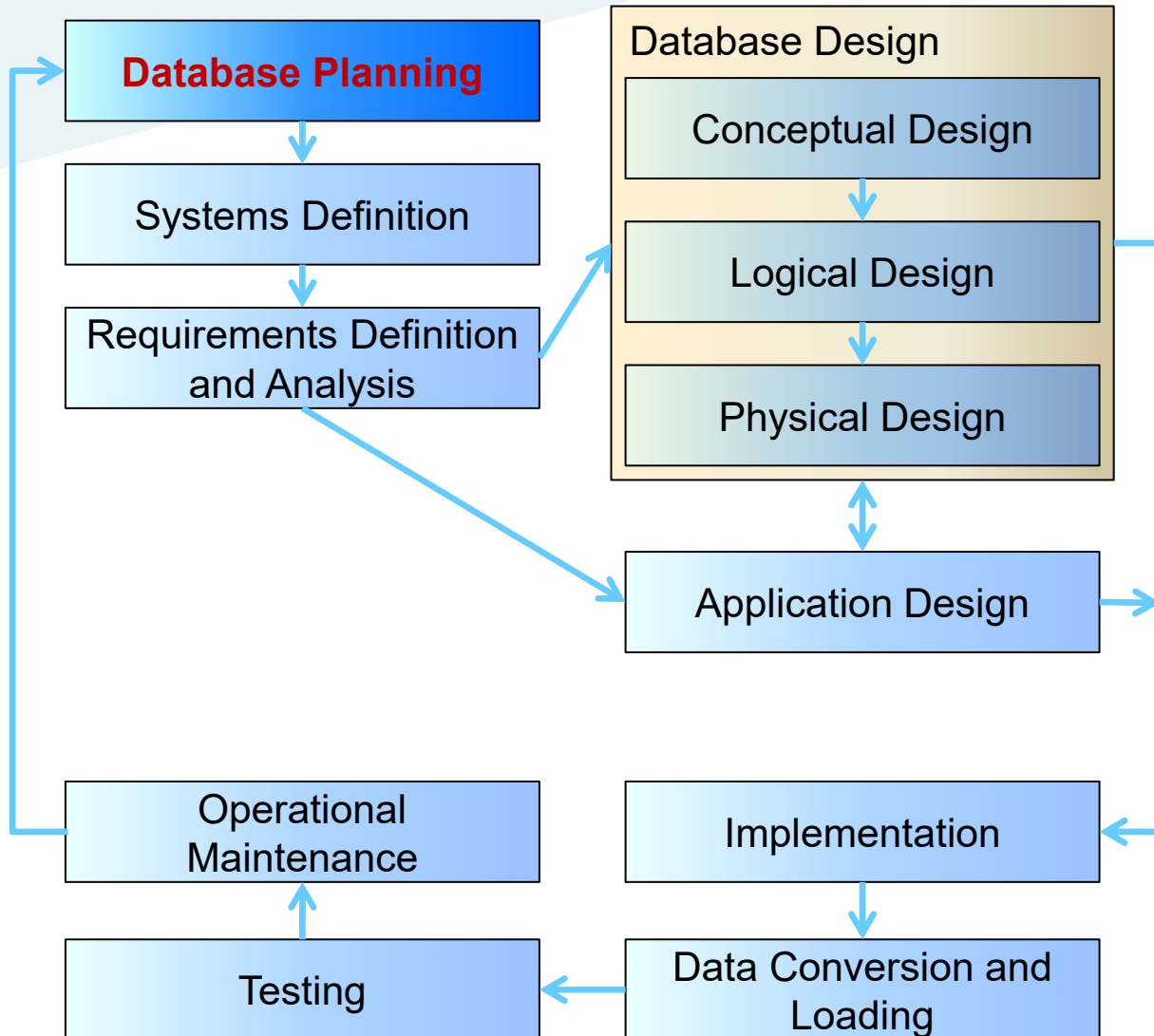
- GRANT
- REVOKE



Database Development Lifecycle

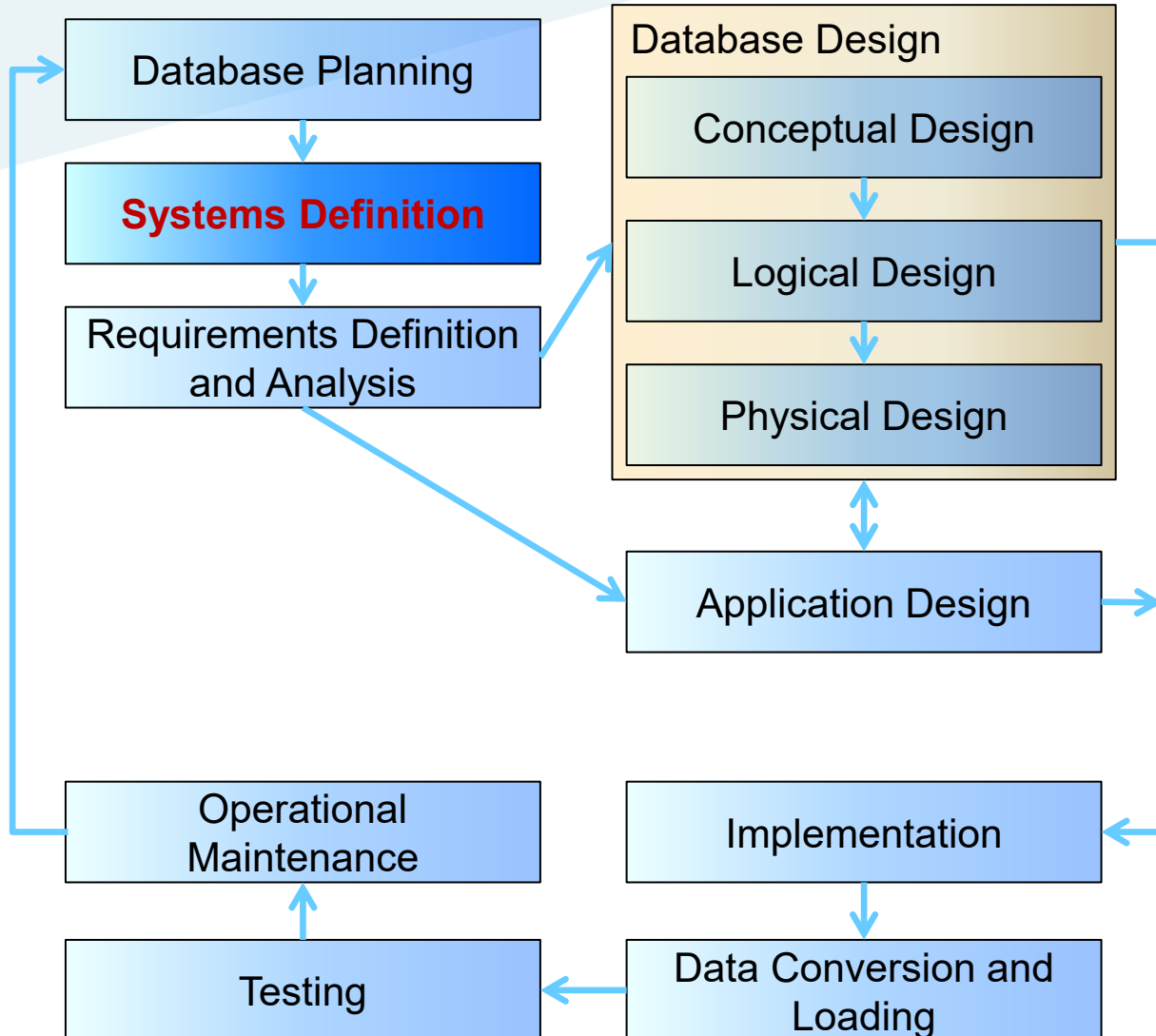
Part of system development lifecycle

Database Development Lifecycle



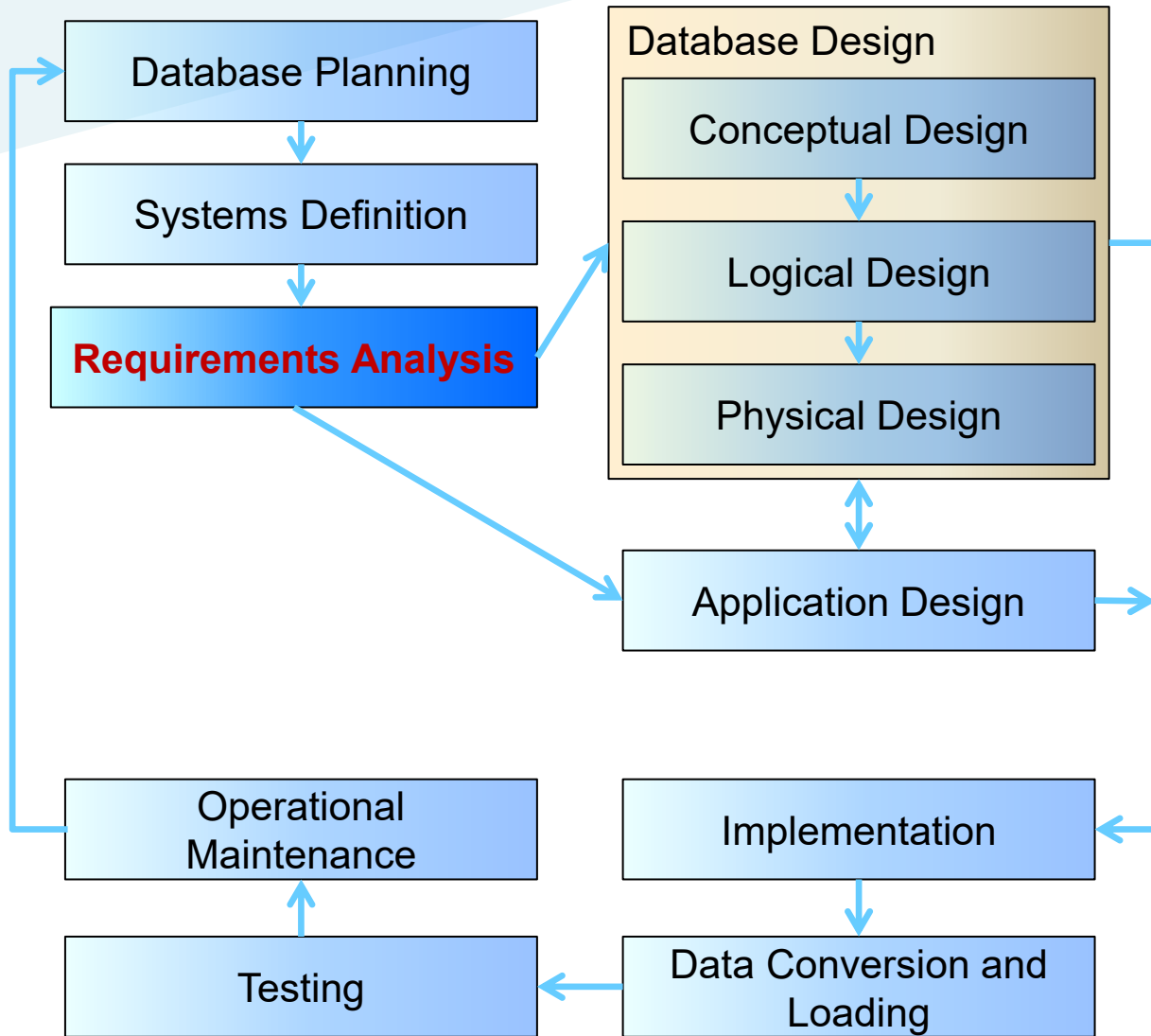
- Planning how to do the project.
 - How does the enterprise work
 - Enterprise data model
- How can the stages be completed efficiently and effectively.
- Outside scope of the course

Database Development Lifecycle



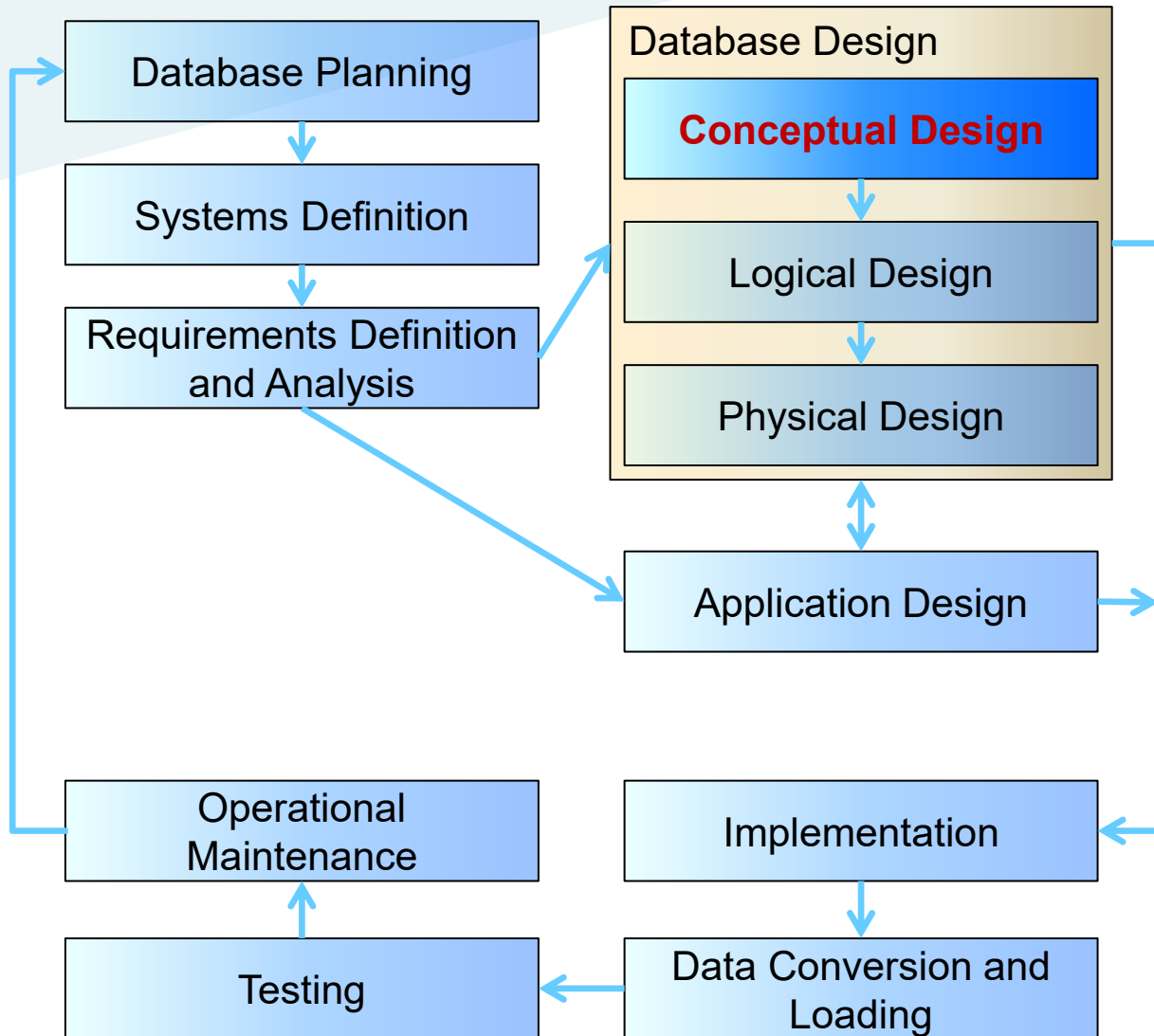
- Specifying scope and boundaries
 - Users
 - Major user views
 - Application areas
- How does it interact with other systems
- User views – how the system operates from differing perspectives
- Outside scope of the course (slightly)

Database Development Lifecycle



- Collection and analysis of requirements for the new system

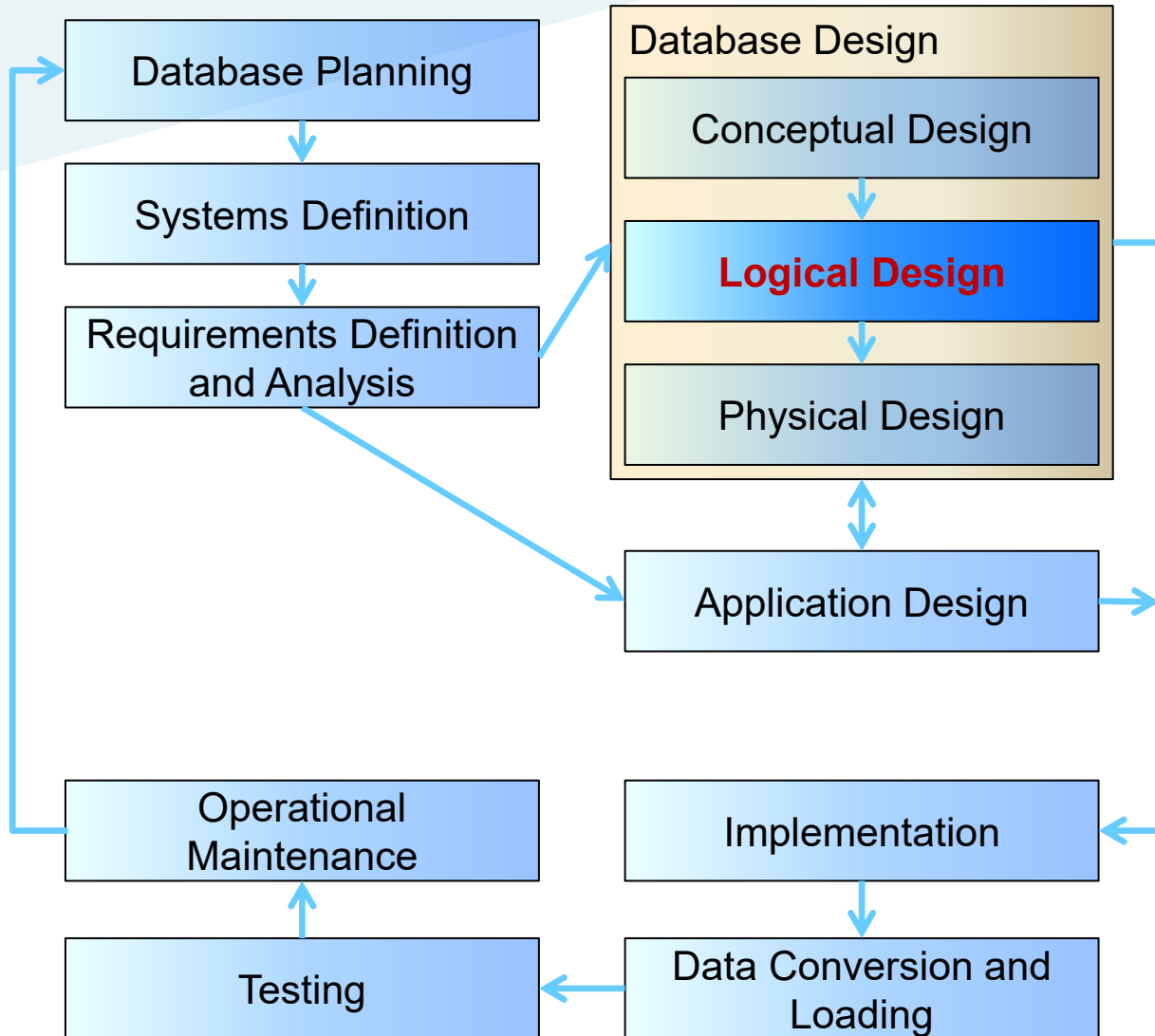
Database Development Lifecycle



- High-level, first-pass model of entities and their connections
- Typically omits attributes*
- Could potentially be implemented in a non-relational database
- Thus can include many-to-many relationships, repeating groups, composite attributes

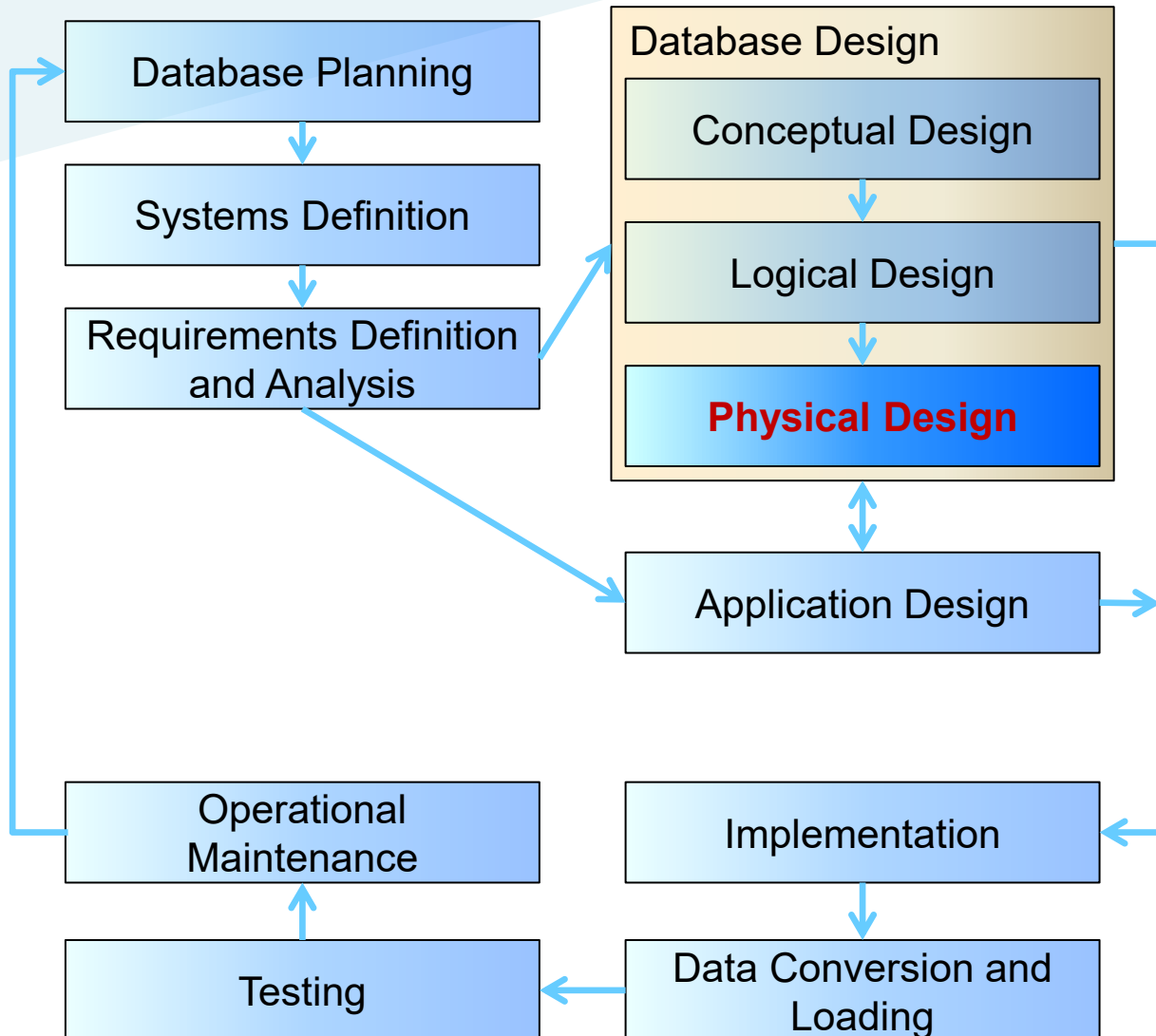
* Typically we list only attributes in the case study

Database Development Lifecycle



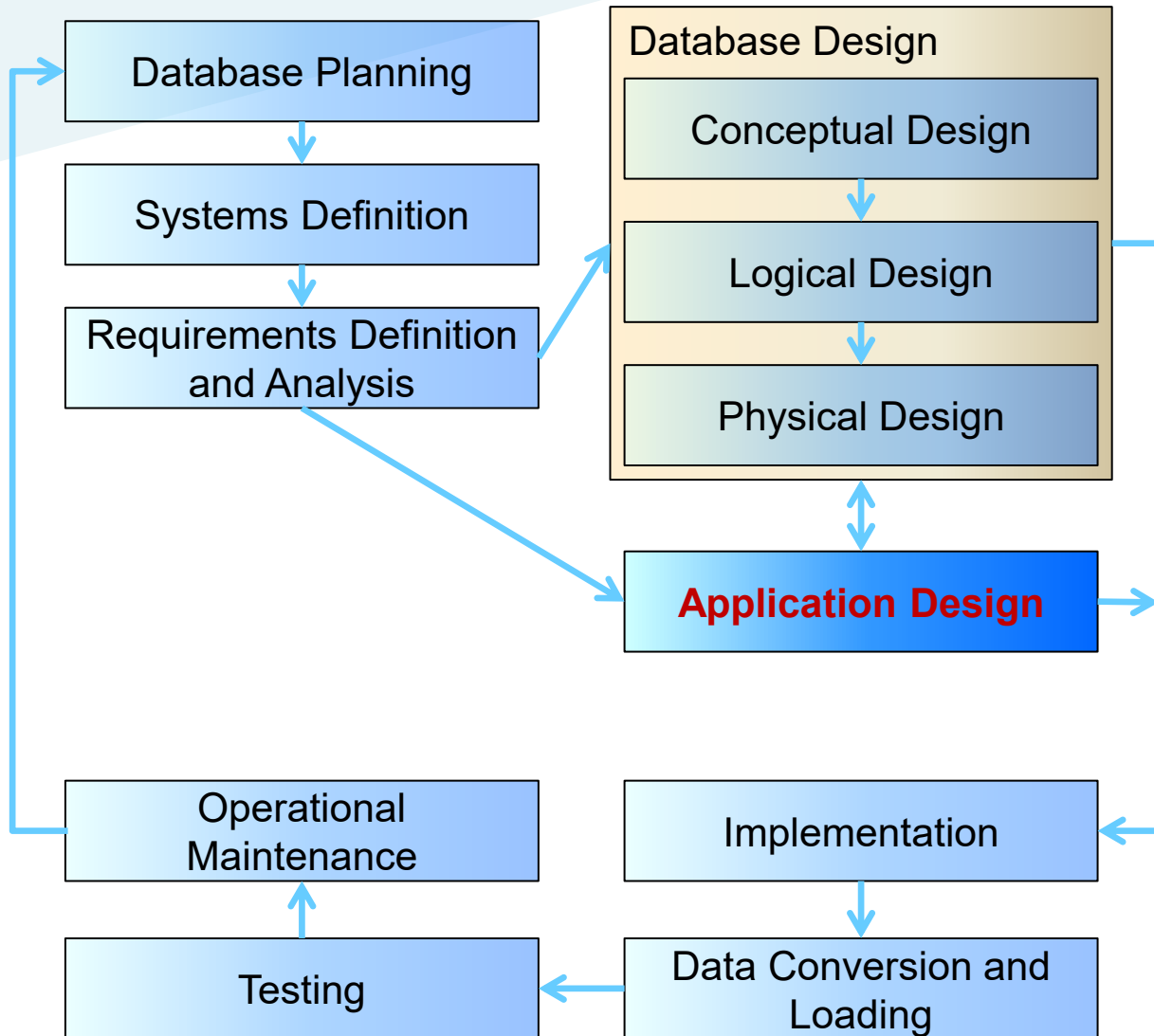
- Builds on the conceptual design
- Designing now for a relational database
- Includes columns and keys
- Independent of a specific vendor and other physical considerations

Database Development Lifecycle



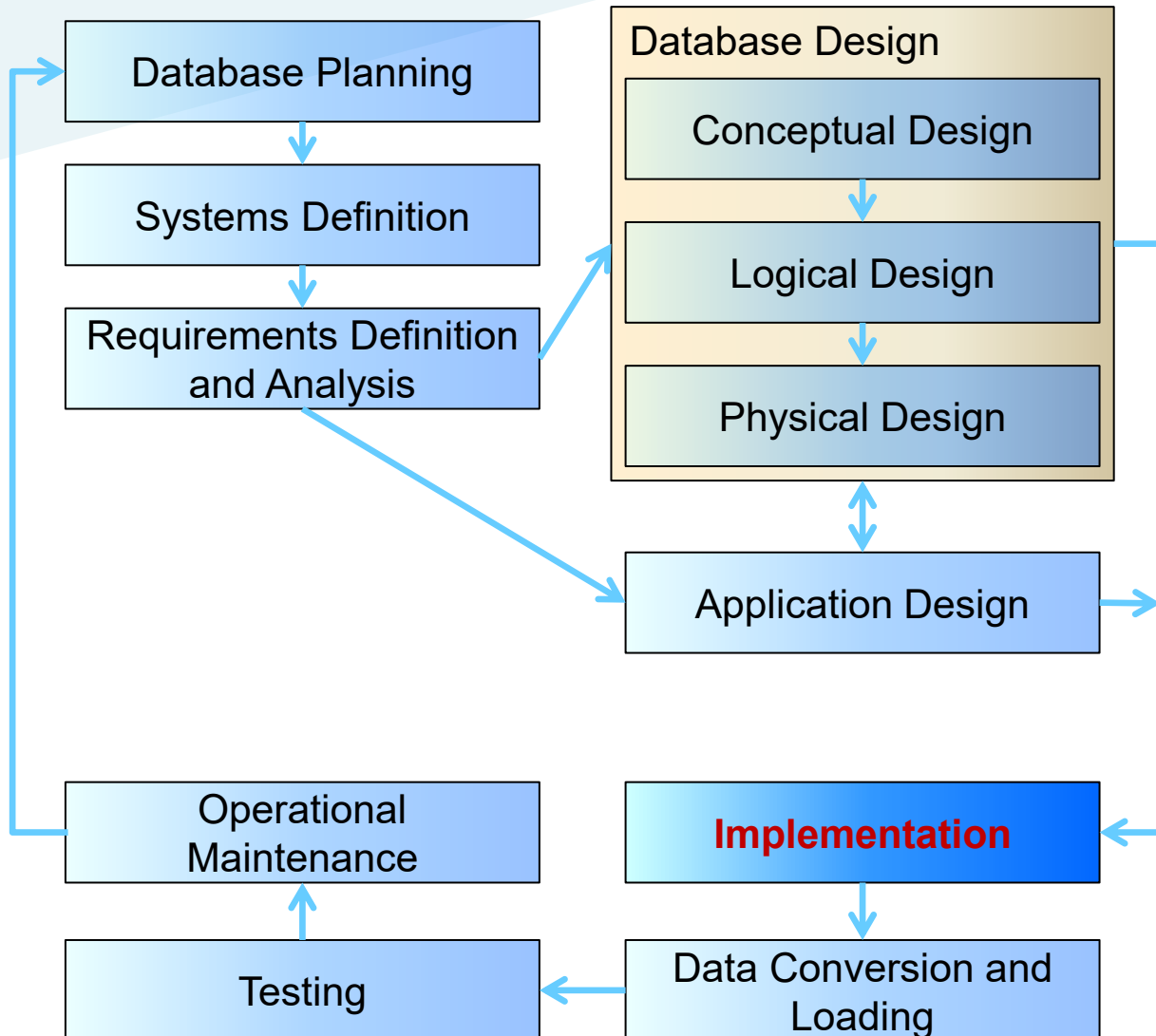
- Implements the logical design for a specific DBMS.
- Describes:
 - Base tables
 - Data types
 - Indexes
 - Integrity constraints
 - File organisation
 - Security measures
- We will cover some aspects of physical design

Database Development Lifecycle



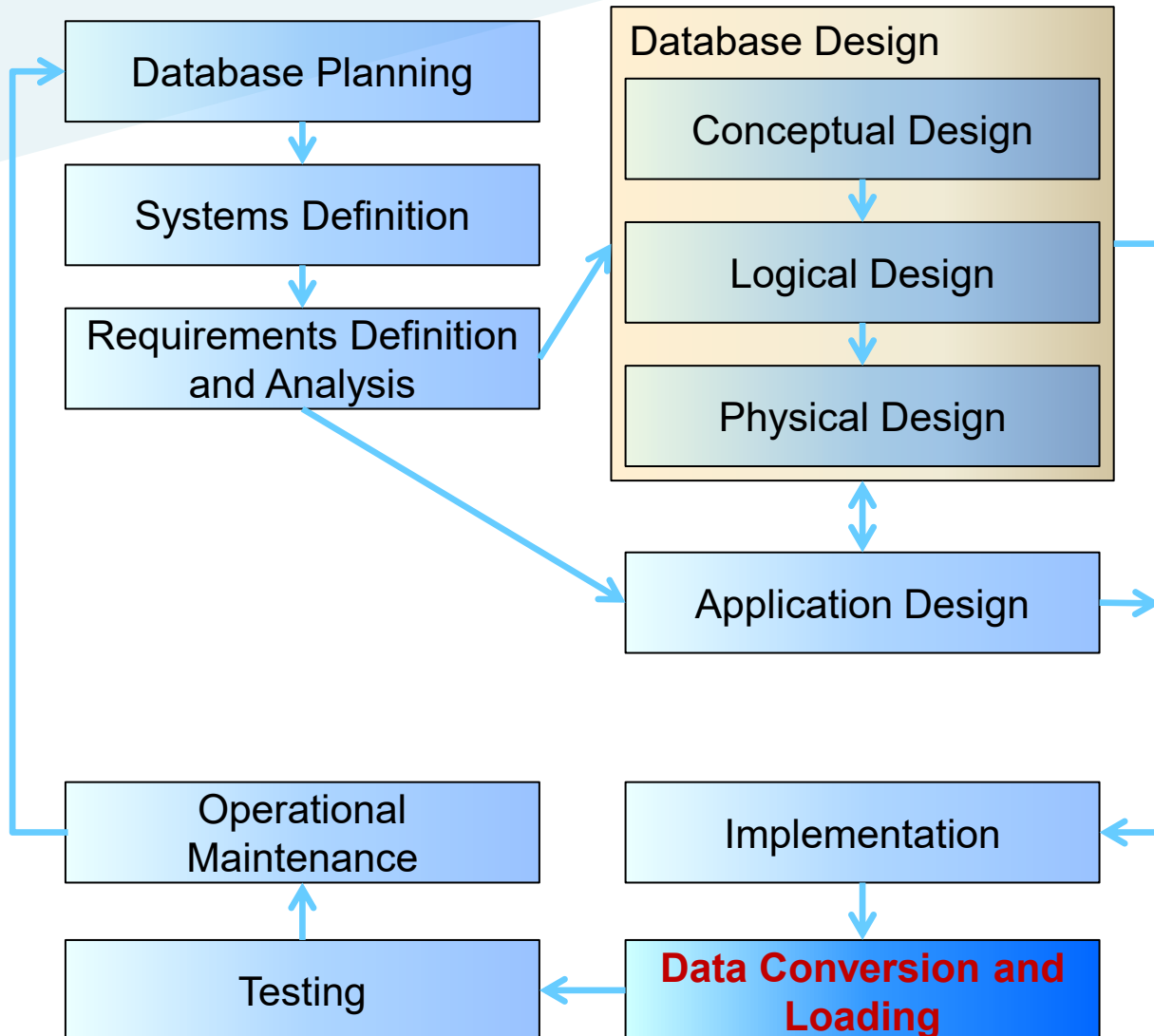
- Done in conjunction with database design
- Design of the interface and application programs that use and process the database
- Mostly outside scope of the course, but briefly discussed in the Web Apps lecture

Database Development Lifecycle



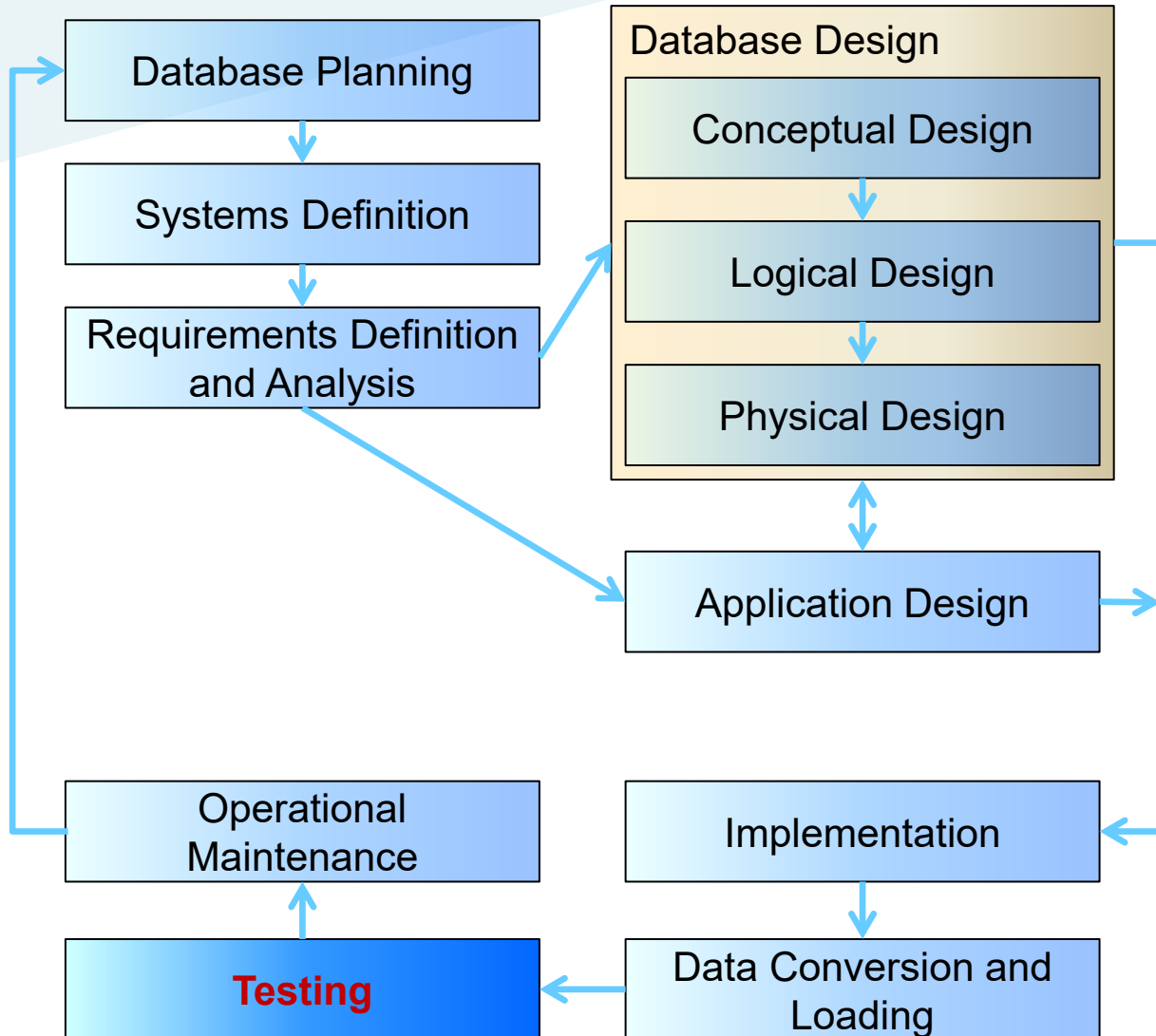
- Implementation of the design as a working database

Database Development Lifecycle



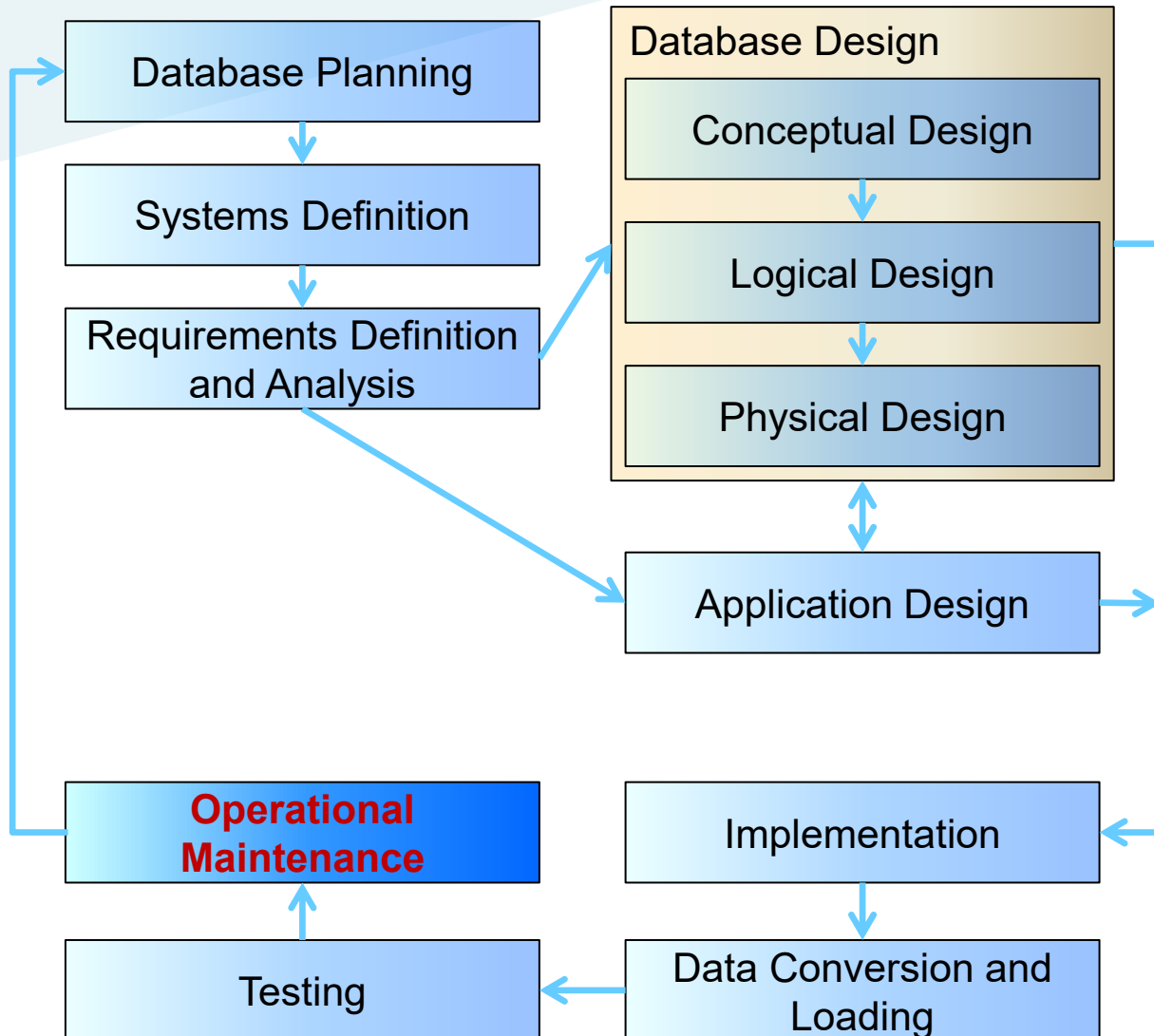
- Transfer existing data into the database
- Conversion from old systems
- Non trivial task
- Mostly outside scope of the course (concepts covered in the Data Warehouse lecture)

Database Development Lifecycle



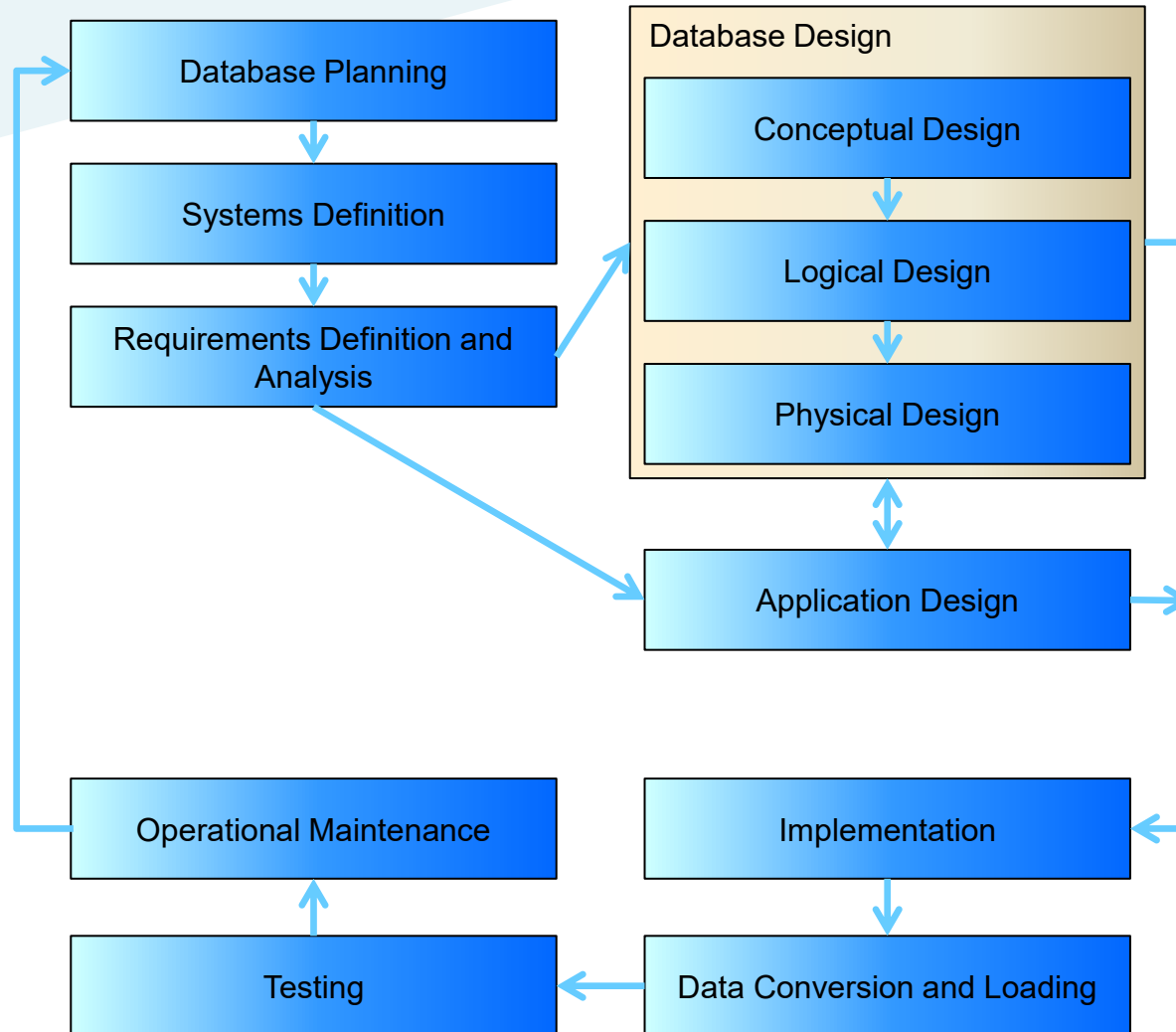
- Running the database to find errors in the design / setup
- Other issues also
 - Performance
 - Robustness
 - Recoverability
 - Adaptability
 - Security
- Mostly outside scope of the course (see ISYS90086 Data Warehousing)

Database Development Lifecycle



- The process of monitoring and maintaining the database following its commissioning
- Monitoring and improving performance
- Handling changes to requirements
- We will touch on some of these topics later in lectures 15, 20

Database Lifecycle (Summary)





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